C4-3b: Normalization

Database Concept

Normalization

Normalisation is the process of organising the tables in a database to reduce data redundancy and prevent inconsistent data.

For GCE A Level syllabus, we will only focus on three normal forms:

- first normal form (1NF)
- second normal form (2NF)
- third normal form (3NF).

First Normal Form (1NF)

For a table to be in 1NF, all the columns must be atomic.

MatricNo	Name	Gender	CivicsClass	CivicsTutor	HomeRoom	CCAInfo
1	Adam	M	18S12	Peter Lim	TR1	Tennis Teacher IC = Adrian Tan
2	Adrian	M	18S12	Peter Lim	TR1	Choir Teacher IC = Adeline Wong, Student Council Teacher IC = David Leong
3	Adam	M	18A10	Pauline Lee	TR2	Rugby Teacher IC = Andrew Quah
4	Bala	M	18A10	Pauline Lee	TR2	Badminton Teacher IC = Lilian Lim
5	Bee Lay	F	18A10	Pauline Lee	TR2	Choir Teacher IC = Adeline Wong, Chess Club Teacher IC = Edison Pon

Primary Key = {MatricNo}

This table is not in 1NF because the the field CCAInfo contains multiple values.

First Normal Form (1NF)

In order to make the table to be in 1NF, CCAInfo field can be split to two single-value fields CCAName and CCATeacherIC.

MatricNo	Name	Gender	CivicsClass	CivicsTutor	HomeRoom	CCAName	CCATeacherIC
1	Adam	M	18S12	Peter Lim	TR1	Tennis	Adrian Tan
2	Adrian	M	18S12	Peter Lim	TR1	Choir	Adeline Wong
2	Adrian	M	18S12	Peter Lim	TR1	Student Council	David Leong
3	Adam	M	18A10	Pauline Lee	TR2	Rugby	Andrew Quah
4	Bala	M	18A10	Pauline Lee	TR2	Badminton	Lilian Lim
5	Bee Lay	F	18A10	Pauline Lee	TR2	Choir	Adeline Wong
5	Bee Lay	F	18A10	Pauline Lee	TR2	Chess Club	Edison Poh

Primary Key = {MatricNo,CCAName}

Note: Since the students with MatricNo 2 and 5 have multiple CCAs, we need to split them into two records (rows) so that the values for the field CCAName are atomic.

The Primary Key (PK) will need to be composite {MatricNo,CCAName} to identify each record.

Second Normal Form (2NF)

For a table to be in 2NF, it must satisfy two conditions:

1. The table should be in 1NF
2. Every non-key attribute must be fully dependent on the entire primary key. (i.e. No attribute can depend only on part of the PK.)

MatricNo	Name	Gender	CivicsClass	CivicsTutor	HomeRoom	CCAName	CCATeacherIC
1	Adam	M	18S12	Peter Lim	TR1	Tennis	Adrian Tan
2	Adrian	M	18S12	Peter Lim	TR1	Choir	Adeline Wong
2	Adrian	M	18S12	Peter Lim	TR1	Student Council	David Leong
3	Adam	M	18A10	Pauline Lee	TR2	Rugby	Andrew Quah
4	Bala	M	18A10	Pauline Lee	TR2	Badminton	Lilian Lim
5	Bee Lay	F	18A10	Pauline Lee	TR2	Choir	Adeline Wong
5	Bee Lay	F	18A10	Pauline Lee	TR2	Chess Club	Edison Poh

Primary Key = {MatricNo,CCAName}

Second Normal Form (2NF)

Primary Key = {MatricNo,CCAName}

Name, Gender, CivicsClass, CivicsTutor and HomeRoom is dependent only on **MatricNo**.

CCATeacherIC is dependent only on **CCAName**.

Hence, this table is not in 2NF since all attributes are dependent only on part of the primary key, and not solely dependent on the entire primary key.

Second Normal Form (2NF)

To change it to 2NF, we decompose the table into three smaller tables:

Student

MatricNo	Name	Gender	CivicsClass	CivicsTutor	HomeRoom
1	Adam	M	18S12	Peter Lim	TR1
2	Adrian	M	18S12	Peter Lim	TR1
3	Adam	M	18A10	Pauline Lee	TR2
4	Bala	M	18A10	Pauline Lee	TR2
5	Bee Lay	F	18A10	Pauline Lee	TR2

Primary Key = {MatricNo}

StudentCCA

MatricNo	CCAName
1	Tennis
2	Choir
2	Student Council
3	Rugby
4	Badminton
5	Choir
5	Chess Club

Primary Key = {MatricNo, CCAName}

CCAInfo

CCAName	CCATeacherIC
Tennis	Adrian Tan
Choir	Adeline Wong
Student Council	David Leong
Rugby	Andrew Quah
Badminton	Lilian Lim
Chess Club	Edison Poh

Primary Key = {CCAName}

All the tables are 2NF since all the non-key attributes are solely dependent on the primary key.

Foreign Key (FK)

Student

MatricNo	Name	Gender	CivicsClass	CivicsTutor	HomeRoom
1	Adam	M	18S12	Peter Lim	TR1
2	Adrian	M	18S12	Peter Lim	TR1
3	Adam	M	18A10	Pauline Lee	TR2
4	Bala	M	18A10	Pauline Lee	TR2
5	Bee Lay	F	18A10	Pauline Lee	TR2

Primary Key = {MatricNo}

StudentCCA

MatricNo	CCAName
1	Tennis
2	Choir
2	Student Council
3	Rugby
4	Badminton
5	Choir
5	Chess Club

Primary Key = {MatricNo, CCAName}

Foreign Key = {MatricNo*}

CCAInfo

CCAName	CCATeacherIC
Tennis	Adrian Tan
Choir	Adeline Wong
Student Council	David Leong
Rugby	Andrew Quah
Badminton	Lilian Lim
Chess Club	Edison Poh

Primary Key = {CCAName}

Foreign Key = {CCAName*}

The values for the MatricNo field in the StudentCCA table is referencing to the Primary Key (PK) for the Student table.

Hence the MatricNo field is a Foreign Key (FK) in the StudentCCA table, often denoted with an asterisk (*) symbol.

Foreign Key (FK)

Student

MatricNo	Name	Gender	CivicsClass	CivicsTutor	HomeRoom
1	Adam	M	18S12	Peter Lim	TR1
2	Adrian	M	18S12	Peter Lim	TR1
3	Adam	M	18A10	Pauline Lee	TR2
4	Bala	M	18A10	Pauline Lee	TR2
5	Bee Lay	F	18A10	Pauline Lee	TR2

Primary Key = {MatricNo}

StudentCCA

MatricNo	CCAName
1	Tennis
2	Choir
2	Student Council
3	Rugby
4	Badminton
5	Choir
5	Chess Club

Primary Key = {MatricNo, CCAName}

Foreign Key = {MatricNo*}

CCAInfo

CCAName	CCATeacherIC
Tennis	Adrian Tan
Choir	Adeline Wong
Student Council	David Leong
Rugby	Andrew Quah
Badminton	Lilian Lim
Chess Club	Edison Poh

Primary Key = {CCAName}

Foreign Key = {CCAName*}

The values for the CCAName field in the StudentCCA table is referencing to the Primary Key (PK) for the CCAInfo table.

Hence the CCAName field is a Foreign Key (FK) in the StudentCCA table, often denoted with an asterisk (*) symbol.

Third Normal Form (3NF)

For a table to be in 3NF, it must satisfy two conditions:

1. The table should be in 2NF
2. The table should not have transitive dependencies

Student

MatricNo	Name	Gender	CivicsClass	CivicsTutor	HomeRoom
1	Adam	M	18S12	Peter Lim	TR1
2	Adrian	M	18S12	Peter Lim	TR1
3	Adam	M	18A10	Pauline Lee	TR2
4	Bala	M	18A10	Pauline Lee	TR2
5	Bee Lay	F	18A10	Pauline Lee	TR2

For table Student,

{CivicsTutor, HomeRoom} are dependent on CivicsClass

{CivicsClass} is dependent on MatricNo

Therefore, {CivicsTutor, HomeRoom} are transitively dependent on MatricNo.

Third Normal Form (3NF)

To remove the transitive dependency, we decompose the Student table

Student

MatricNo	Name	Gender	CivicsClass
1	Adam	M	18S12
2	Adrian	M	18S12
3	Adam	M	18A10
4	Bala	M	18A10
5	Bee Lay	F	18A10

Primary Key = {MatricNo}
Foreign Key = {CivicsClass*}

Civics

CivicsClass	CivicsTutor	HomeRoom
18S12	Peter Lim	TR1
18A10	Pauline Lee	TR2

Primary Key = {CivicsClass}

The CivicsClass field is a Foreign Key (FK) in the Student table referencing to the Primary Key (PK) CivicsClass in the Civics table.

The Normalised Tables

Student

MatricNo	Name	Gender	CivicsClass
1	Adam	M	18S12
2	Adrian	M	18S12
3	Adam	M	18A10
4	Bala	M	18A10
5	Bee Lay	F	18A10

Primary Key = {MatricNo}
Foreign Key = {CivicsClass}

StudentCCA

MatricNo	CCAName
1	Tennis
2	Choir
2	Student Council
3	Rugby
4	Badminton
5	Choir
5	Chess Club

Primary Key = {MatricNo, CCAName}
Foreign Key = {MatricNo}
Foreign Key = {CCAName}

CCAInfo

CCAName	CCATeacherIC
Tennis	Adrian Tan
Choir	Adeline Wong
Student Council	David Leong
Rugby	Andrew Quah
Badminton	Lilian Lim
Chess Club	Edison Poh

Primary Key = {CCAName}

Table Descriptions:

Student (MatricNo, Name, Gender, CivicsClass*)
Civics (CivicsClass, CivicsTutor, HomeRoom)
StudentCCA (MatricNo*, CCAName*)
CCAInfo (CCAName, CCATeacherIC)

PK denoted by underline and FK by *

Civics

CivicsClass	CivicsTutor	HomeRoom
18S12	Peter Lim	TR1
18A10	Pauline Lee	TR2

Primary Key = {CivicsClass}

The Normalised Tables

Student

MatricNo	Name	Gender	CivicsClass
1	Adam	M	18S12
2	Adrian	M	18S12
3	Adam	M	18A10
4	Bala	M	18A10
5	Bee Lay	F	18A10

Primary Key = {MatricNo}
Foreign Key = {CivicsClass}

StudentCCA

MatricNo	CCAName
1	Tennis
2	Choir
2	Student Council
3	Rugby
4	Badminton
5	Choir
5	Chess Club

Primary Key = {MatricNo, CCAName}
Foreign Key = {MatricNo}
Foreign Key = {CCAName}

CCAInfo

CCAName	CCATeacherIC
Tennis	Adrian Tan
Choir	Adeline Wong
Student Council	David Leong
Rugby	Andrew Quah
Badminton	Lilian Lim
Chess Club	Edison Poh

Primary Key = {CCAName}

Table Descriptions:

Student (MatricNo, Name, Gender, CivicsClass)
Civics (CivicsClass, CivicsTutor, HomeRoom)
StudentCCA (MatricNo, CCAName)
CCAInfo (CCAName, CCATeacherIC)

PK denoted by underline and FK by dotted underline.

Civics

CivicsClass	CivicsTutor	HomeRoom
18S12	Peter Lim	TR1
18A10	Pauline Lee	TR2

Primary Key = {CivicsClass}